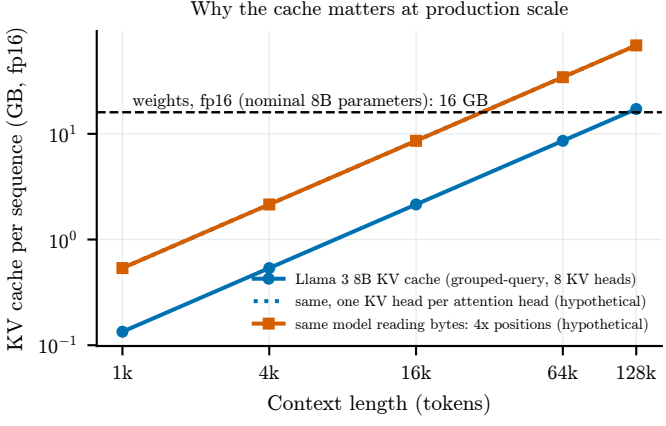
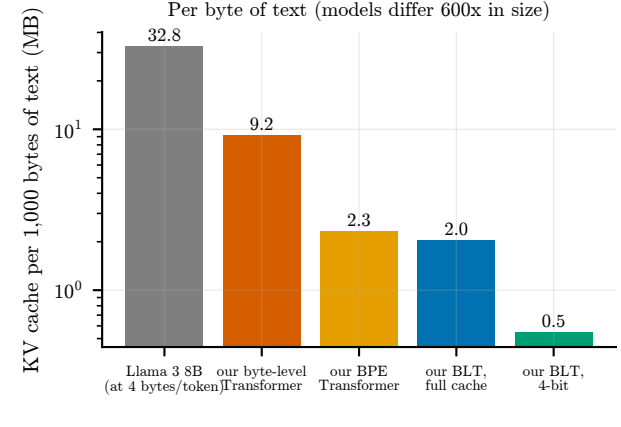

Why the cache matters at production scale
weights, fp16 (nominal 8B parameters): 16 GB
KV cache per sequence (GB, fp16)
10^1
10^0
10^{-1}
Llama 3 8B KV cache (grouped-query, 8 KV heads)
same, one KV head per attention head (hypothetical)
same model reading bytes: 4x positions (hypothetical)
1k
4k
16k
64k
128k
Context length (tokens)


Per byte of text (models differ 600x in size)
KV cache per 1,000 bytes of text (MB)
10^1
10^0
32.8
9.2
2.3
2.0
0.5
Llama 3 8B (at 4 bytes/token)
our byte-level Transformer
our BPE Transformer
our BLT, full cache
our BLT, 4-bit